IMDb: Decoding What Makes a Movie Timeless

An Analytical Study of IMDb Top 700 Movies

DATA ANALYTICS GRADUATION PROJECT

Python|pandas|Tableau |Google Colab

EXECUTIVE SUMMARY

This project analyses IMDb's Top 700 movies to understand the patterns associated with high and lasting movie reputation. IMDb Rating is used as the primary performance measure, while genre, era, votes, runtime, director, actor and country representation are examined to identify meaningful patterns within the dataset.

• DATA SCOPE & DATASET

- SCOPE OF ANALYSIS

This analysis examines the patterns associated with IMDb ratings across the top 700 movies. It evaluates movie performance across genres and eras, investigates the relationship between votes, runtime and IMDb rating, and examines director and actor appearance patterns. Country representation is also analyzed to understand the composition of the dataset. Statistical analysis is used to distinguish meaningful relationships from simple patterns

- THE DATA

The dataset contains **700 IMDb Top movies** and **10 analytical fields** covering movie characteristics, audience ratings, popularity, creative contributors and country information. **Average IMDb Rating** is used as the primary performance metric, supported by votes, runtime, director frequency and country representation.

700	**10**	**8.00**	**533K**	**127.4 min**
Movies	Fields	Avg Rating	Avg Votes	Avg Runtime

DATASET SCHEMA

Field	Description
Rank	Movie's position within the Top 700
Title	Movie title
Year	Movie release year
Genre(s)	Genre information provided for the movie
Director	Director associated with the movie
Main Actor(s)	Main actor information
Country	Country information provided in the dataset
IMDb Rating	IMDb user rating
Votes	Number of IMDb votes
Runtime (mins)	Movie duration in minutes

DATA CLEANING & PREPROCESSING

1. DATA QUALITY CHECKS

Before conducting exploratory and statistical analysis, the IMDb dataset was systematically checked for duplicates, missing values, data types, invalid values and potential outliers. These checks were performed to ensure that the analysis was based on consistent and reliable records.

2. DUPLICATE CHECK

No complete duplicate records were identified. A secondary check using the combination of Movie Title and Release Year also found no duplicate movie records. Therefore, no duplicate rows were removed.

Complete Duplicates: 0

Title-Year Duplicates: 0

3. MISSING VALUES

The core analytical fields, including IMDb Rating, Votes, Year and Runtime, contain no missing values. Missing values were identified only in the Main Actor(s) field for 10 movies. Since actor information was not required for every analysis, these records were retained rather than removed.

Missing Main Actor(s): 10

4. TRANSFORMATION & OUTLIER HANDLING

Genre(s) and Country were transformed into individual values for representation analysis because several movies contain multiple genres or countries. Runtime was also checked using the IQR method. Fourteen high-runtime observations were identified as statistical outliers; however, these values represented genuine long-duration movies and were therefore retaine

14 Runtime Outliers → Retained as Genuine Observations

CLEANING PRINCIPLE

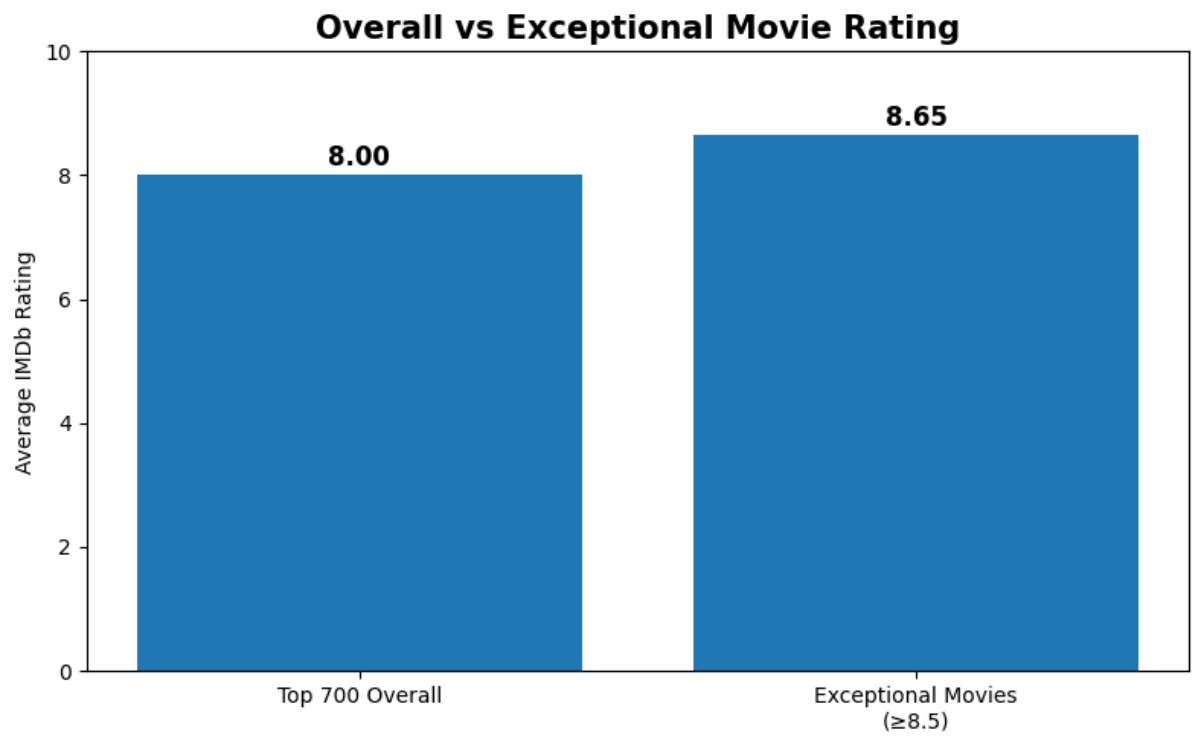
No valid movie records were removed solely because they were statistically unusual. Transformations were applied only where necessary to support accurate analysis.

EXPLORATORY DATA ANALYSIS (EDA)

OVERALL RATING BENCHMARK

Across the 700 movies, the average IMDb Rating is 8.00, establishing a high overall rating benchmark for the dataset. To identify exceptional performers, movies with an IMDb Rating of 8.5 or above were examined separately. This group contains 62 movies, representing 8.86% of the Top 700, with an average rating of 8.65.

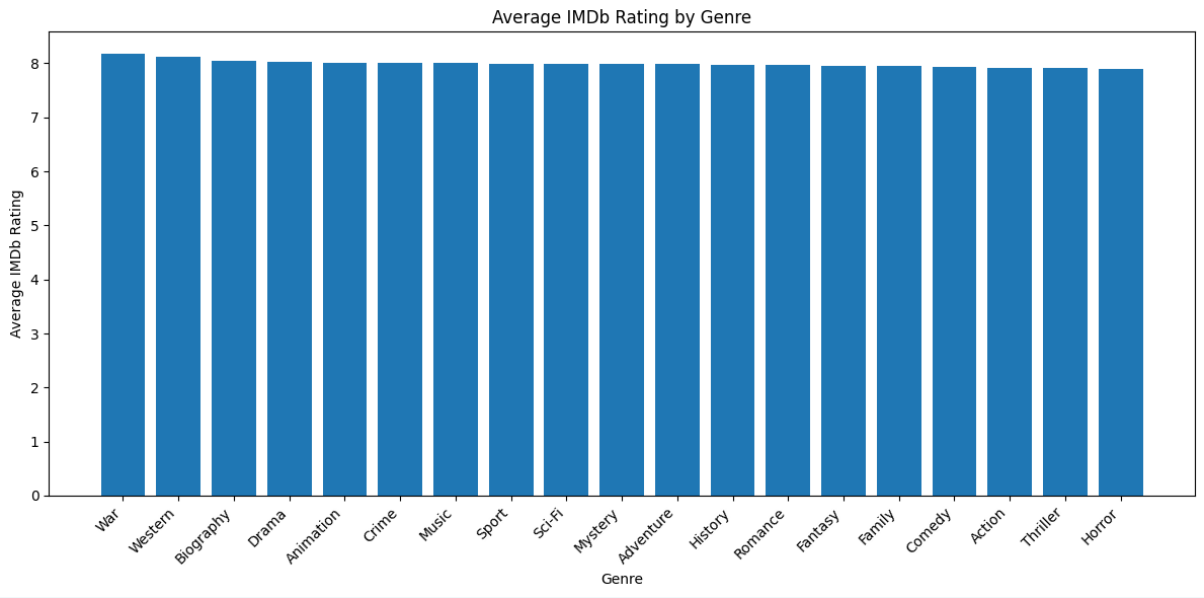
62 Movies | 8.86% of top 700 | 8.65 Avg Rating



GENRE PERFORMANCE ANALYSIS

1. GENRE-LEVEL PERFORMANCE

To understand how movie performance varies across different genres, the analysis compares the Average IMDb Rating for each genre.



Since movies can belong to multiple genres, the genre information was split into individual genre categories before calculating these metrics. Genres with at least 10 movie appearances were used for the main rating comparison to reduce the influence of very small samples.

KEY FINDINGS

• War has the highest average IMDb Rating among the displayed genres at approximately 8.18, followed by Western (8.12) and Biography (8.04).

• Drama is the most represented genre, with 473 movie appearances, and has an average rating of approximately 8.03.

• Horror, Thriller and Action appear toward the lower end of the rating comparison, at approximately 7.89, 7.92 and 7.92, respectively.

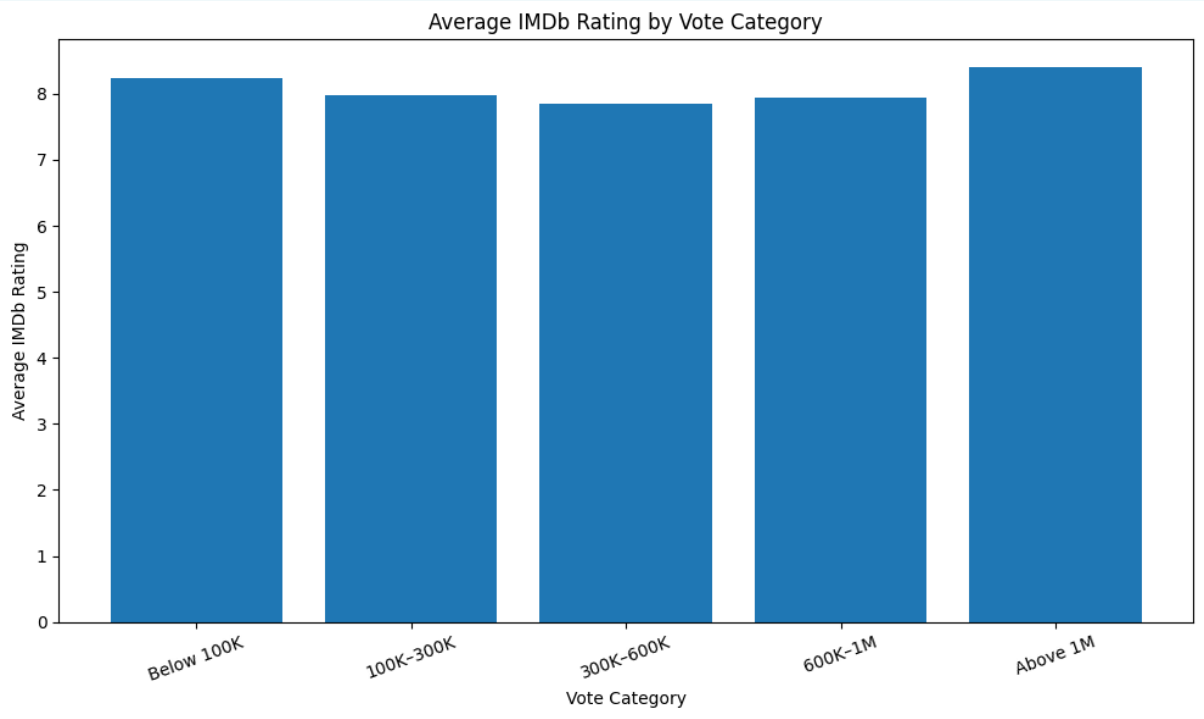
• The differences between most genres are relatively small, with average ratings concentrated around the 8.0 level.

AUDIENCE ENGAGEMENT & IMDb RATING

1. IMDb RATING BY VOTE CATEGORY

To understand how audience engagement varies across movies, the analysis compares Average IMDb Rating across different Vote Categories.

The number of Votes represents the level of audience engagement received by a movie. Movies were grouped into vote-volume categories, and the Average IMDb Rating was calculated for each category to identify whether higher audience engagement is consistently associated with higher ratings.



KEY FINDINGS

• Movies with more than 1M votes have the highest average IMDb Rating at approximately 8.40.

• Movies with fewer than 100K votes also show a relatively high average rating of approximately 8.23.

• The 300K–600K vote category has the lowest average IMDb Rating at approximately 7.85.

• The results do not show a simple pattern where increasing vote volume always leads to a higher IMDb Rating.

INTERPRETATION

The analysis shows that IMDb Rating varies across different levels of audience engagement. The highest average rating is observed among movies with more than 1M votes, but movies with fewer than 100K votes also have a relatively high average rating.

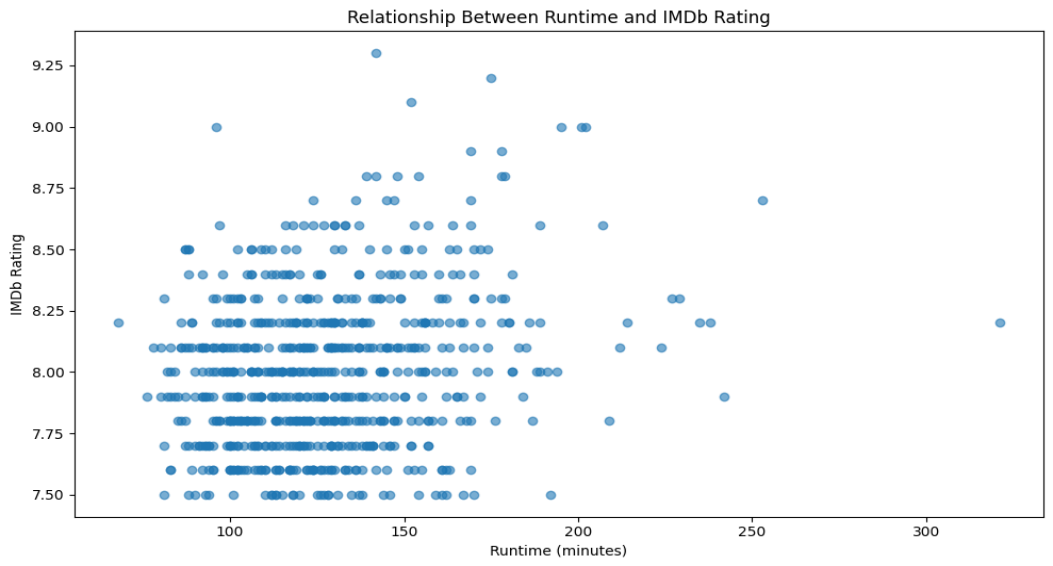
Therefore, higher vote volume should not be interpreted as a direct reason for higher ratings. The relationship appears to be influenced by other characteristics of the movies as well.

BUSINESS INSIGHT

Vote volume can be used as an indicator of audience engagement, but it should not be treated as a standalone measure of movie quality. A stronger evaluation of lasting movie reputation should consider vote volume together with IMDb Rating, genre, era, runtime an

RUNTIME & IMDb RATING

To examine whether movie duration is associated with IMDb Rating, the analysis evaluates the relationship between Runtime and Average IMDb Rating. Runtime was analysed both as a continuous variable and by comparing shorter and longer movies.



Key Findings

Runtime shows a **weak positive relationship** with IMDb Rating (r = **0.239**).

Movies with runtime **above 150 minutes** have an average rating of approximately **8.14**.

Movies with runtime **150 minutes or below** have an average rating of approximately **7.97**.

The difference is statistically significant (**t = 4.92, p < 0.001**).

INTERPRETATION

Longer movies in this dataset tend to have slightly higher IMDb Ratings. However, the correlation is weak, so runtime alone does not explain movie reputation. The statistical test confirms a difference between the two runtime groups, but it should not be interpreted as evidence that longer runtime causes higher ratings.

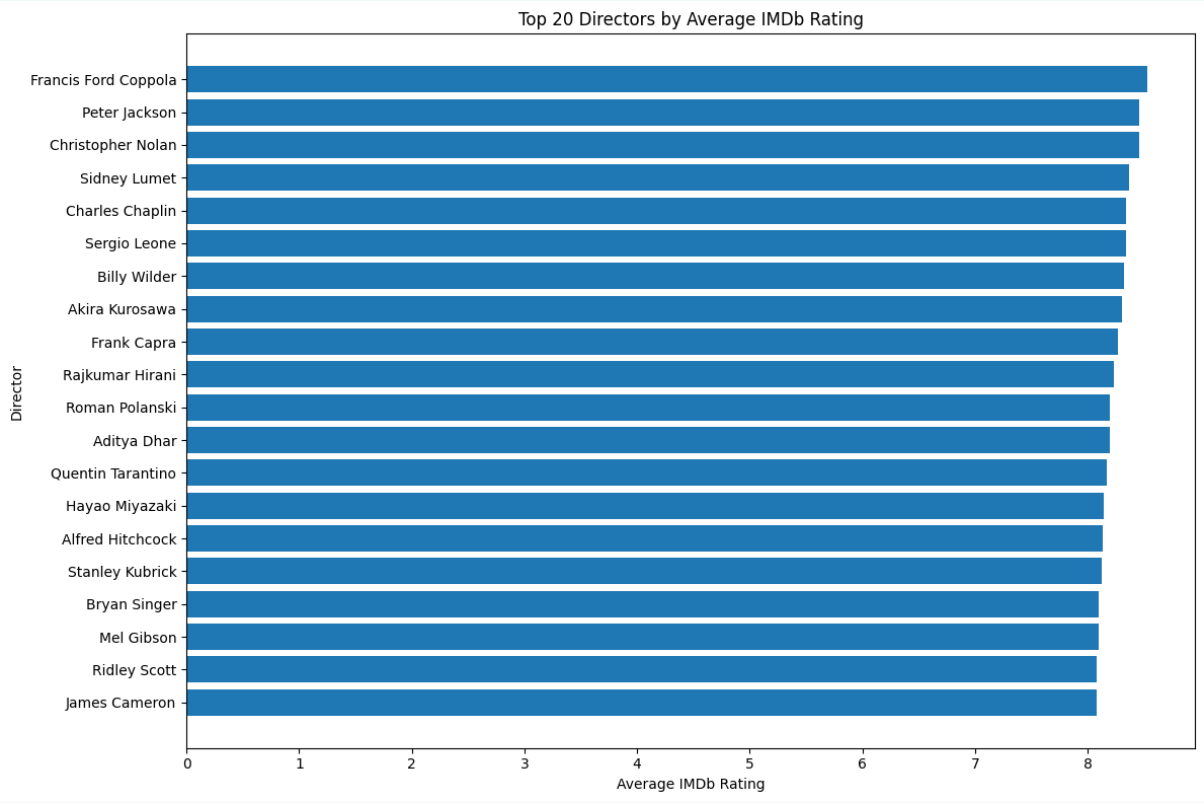
BUSINESS INSIGHT

Runtime can be considered as one supporting characteristic when studying highly rated movies, but it should be evaluated alongside audience engagement, genre, era and other movie-level factors rather than treated as an independent driver of movie quality.

DIRECTOR TRACK RECORD & IMDb RATING

DIRECTOR APPEARANCES IN THE TOP 700

To examine whether repeated appearances in the IMDb Top 700 are associated with stronger movie ratings, directors were evaluated based on their number of appearances and average IMDb Rating



Key Finding

- Francis Ford Coppola: **4 appearances, 8.53 average rating**
- Christopher Nolan: **10 appearances, 8.46**
- Peter Jackson: **5 appearances, 8.46**
- Quentin Tarantino: **10 appearances, 8.17**
- Stanley Kubrick: **10 appearances, 8.12**

Statistical insight

Director appearance frequency and average IMDb Rating show a very weak correlation of approximately 0.064.

Interpretation

Repeated representation in the Top 700 does not automatically correspond to higher average ratings. Some directors with fewer appearances achieve very high average ratings, suggesting that appearance frequency alone is not a strong indicator of movie reputation.

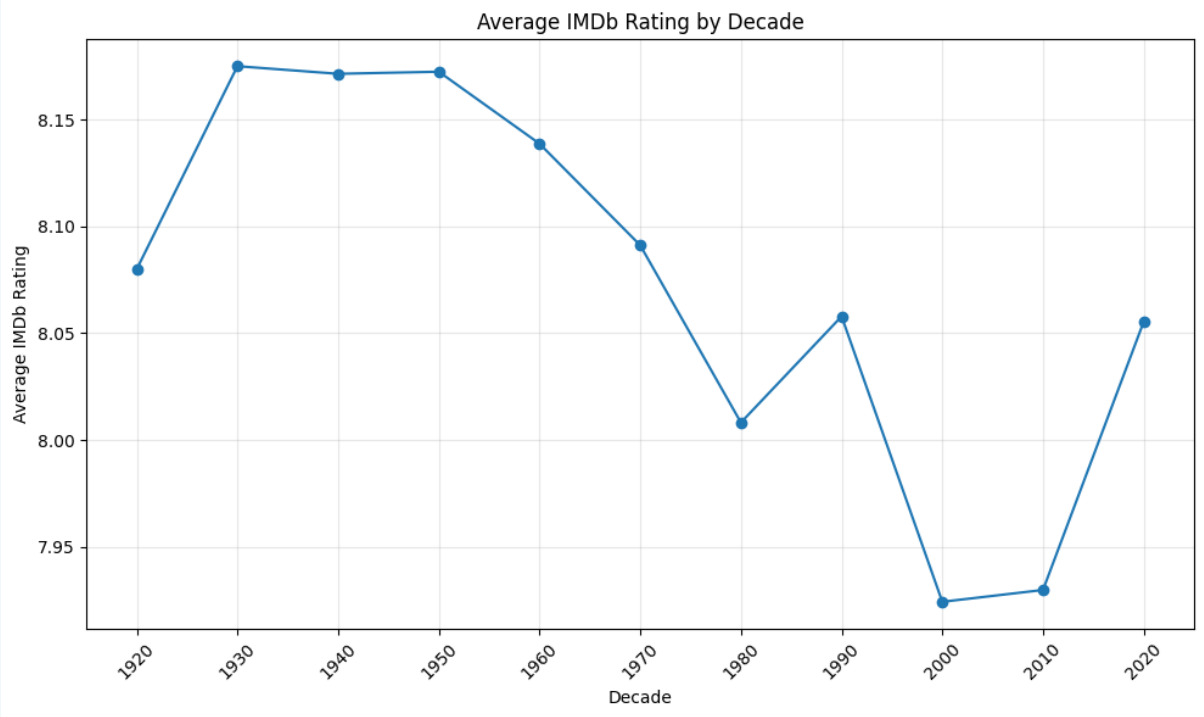
Business Insight

Director track record can provide useful context when evaluating cinematic reputation, but the number of Top 700 appearances should not be treated as a standalone measure of director performance.

ERA & IMDb RATING

IMDb RATING ACROSS MOVIE ERAS

To examine how movie reputation varies across different time periods, the Top 700 movies were grouped into release-era categories and their Average IMDb Rating was compared. This analysis helps identify whether older or newer movies show different rating patterns within the dataset.



KEY FINDINGS

Pre-1980 movies recorded the highest average rating at approximately **8.14.**

The **2000–2009** era recorded the lowest average rating at approximately **7.92.**

The **2010–2019** era showed a similar average rating of approximately **7.93.**

Movies from 2020 onwards showed an increase to approximately **8.06**

INTERPRETATION

The analysis shows that Average IMDb Rating varies across different movie eras. Older movies in the dataset generally show higher average ratings, while movies from the 2000s and 2010s show comparatively lower averages. However, these differences represent patterns within the Top 700 sample and should not be interpreted as evidence that a particular era causes higher or lower movie ratings.

BUSINESS INSIGHT

Era provides useful context when studying lasting movie reputation. The relatively strong ratings observed across several older periods suggest that highly regarded films from earlier eras remain strongly represented in the Top 700. However, era alone is not sufficient to explain movie reputation.

STATISTICAL ANALYSIS

CORRELATION & REGRESSION ANALYSIS

To distinguish meaningful relationships from simple visual patterns, correlation and regression analysis were performed on key movie-level variables. The analysis focused on IMDb Rating, Votes and Runtime to evaluate the strength and statistical significance of their relationships.

Key Statistical Results

Votes ↔ IMDb Rating
Pearson r = 0.405
Moderate positive association.

Runtime ↔ IMDb Rating
Pearson r = 0.239
Weak positive association.

Director Appearances ↔ Average Rating
r = 0.064
Very weak relationship.

Regression Analysis

A multiple regression model was used to evaluate IMDb Rating using Log (Votes) and Runtime as explanatory variables.

R^2 = 0.075
Adjusted R^2 = 0.073
Model p-value < 0.001

INTERPRETATION

Although the regression model is statistically significant, the R^2 of 0.075 indicates that Votes and Runtime explain only a small proportion of the variation in IMDb Rating. This suggests that movie reputation is influenced by additional factors beyond these variables.

KEY INSIGHT

Statistical significance does not imply strong explanatory power. The results show that some relationships are statistically detectable, but their practical strength remains limited.

KEY FINDINGS & BUSINESS INSIGHTS

1. EXCEPTIONAL MOVIES

Only **62 movies (8.86%)** of the Top 700 achieve an IMDb Rating of **8.5 or above**, with an average rating of **8.65**, compared with the overall dataset average of **8.00**.

Insight: Exceptional movies represent a relatively small segment of the dataset and can be treated as a benchmark group for studying lasting movie reputation.

2. AUDIENCE ENGAGEMENT

Movies with more than **1M votes** have the highest average rating at approximately **8.40**, but vote volume does not show a consistent increase in rating across all vote categories.

Insight: Audience engagement provides useful context, but vote volume alone should not be treated as a measure of movie quality.

3. RUNTIME

Runtime has a **weak positive relationship** with IMDb Rating (**r = 0.239**). Movies longer than 150 minutes have an average rating of approximately **8.14**, compared with **7.97** for shorter movies.

Insight: Runtime may be associated with rating, but the relationship is too weak to consider runtime a standalone driver of movie reputation.

4. DIRECTOR TRACK RECORD

Director appearance frequency has a **very weak relationship** with average IMDb Rating (**r = 0.064**).

Insight: A director appearing frequently in the Top 700 does not automatically mean that their movies have higher average ratings.

5. ERA

Pre-1980 movies show the highest average rating at approximately **8.14**, while the **2000–2009** and **2010–2019** periods are around **7.92–7.93**.

Insight: Era provides useful context for understanding lasting movie reputation, but the observed differences should not be interpreted causally.

CORE CONCLUSION

No single variable explains why a movie achieves lasting recognition on IMDb. Movie reputation appears to be a multi-factor outcome involving audience engagement, movie characteristics, creative track record and historical context.

CONCLUSION

This project analysed 700 IMDb Top movies to understand the patterns associated with high and lasting movie reputation.

The analysis shows that no single factor explains IMDb movie reputation. Audience engagement, runtime, genre, era and creative track record provide different perspectives, but their individual relationships with IMDb Rating remain limited.

Statistical analysis further supports this finding. Although some relationships are statistically significant, the regression model explains only **7.5% of the variation in IMDb Rating**, indicating that several other factors contribute to movie reputation.

FINAL TAKEAWAY

A movie becomes timeless not because of one measurable factor, but through the combined influence of **audience engagement, movie characteristics, creative contribution and historical context**.

PROJECT OUTPUT

Python & Pandas → Data Cleaning & Statistical Analysis
Google Collab → Reproducible Analytical Workflow
Tableau → Interactive Dashboard & Visual Storytelling